To turn their vision (pun intended!) into a product, the founders did hardware product design along with standardising their algorithms and paradigms. “One of the taglines that we write on our walls is: Hardware is a part of the algorithm,” says Nikhil. Today, CynLr has a dedicated hardware stack and software layer. On the hardware side, the company is the only one providing depth perception system in the world that has robotic eye movement, convergence, divergence, precision control, etc, that’s enabled through hardware.

“For our customers, we provide the whole system—a robotic arm equipped with vision with

CynLr’s Visual Robot

complete sophisticated grasping capabilities, so that it can handle objects,” says Gokul. Their tech can be applied in part tending, part assembly, and part inspection. The company is open to partnerships—both application partnerships and technology partnerships. Moreover, CynLr is actively hiring for multiple roles. In the near future, CynLr plans to step up its commercialisation and expand abroad. Apart from the automobile industry, they also plan to expand to other domains like logistics and electronics assembly.

—Aaryaa Padhyegurjar

2 NIMAYA: ROBOTS THAT ARE TRANSFORMING THERAPY FOR THE AUTISTIC

In Sanskrit, Nimaya means ‘to create a change,’ which is exactly what this Chennai based startup, Nimaya Innovations, is doing. Founded in 2018, the company has made it its mission to help autistic children become more independent using IoT based robots

Children with autism and multiple disabilities are more attracted to robots and robot-like features. That’s because robots tend to have a routine. For any task, robots will do the same thing repeatedly, a million times over, using the exact same method. Children respond to that well. This is what Dr Ramya Moorthy, Founder of Nimaya Innovations, realised as she went to about 75 special-needs schools and worked with teachers, special educators, and special organisations.

As a part of her PhD, Dr Moorthy designed about five different robots that can assist children in developing daily life skills and their associated psychomotor and cognitive skills. “During my research, I found a lot of studies that have been done on facial expression, social interaction, and communication. But one of the biggest areas that has not been concentrated upon is psychomotor skills,” says Dr Moorthy.

Joystick skill training unit (Credit: Nimaya Innovations)

The company uses unique methods like ‘active learning’ and Suprayoga methodology. Unlike traditional therapy, Nimaya’s robots provide children with audio-visual feedback, which enables them to learn an activity quickly and also retain it for a longer period of time. Most importantly, children are able to generalise an activity to everyday life. Suppose a teacher shows children an apple and says,